

Problematic Internet Usage in Youths: An Ethical Audit of an ADS

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Abstract

The internet has fundamentally changed how people interact with one another and the world. Automated decision systems, such as ones created for the Child Mind Institute’s Problematic Internet Use Kaggle competition, explore potential predictions for diagnoses of certain disorders. In this paper we will conduct an audit on the #1 model from the competition to determine its performance capabilities. We will later discuss these findings and what they indicate moving forward.

1 Introduction

The effect of the digital world on children’s well-being has become a hot topic in the medical and mental health sectors. Growing concern over harmful internet use on this vulnerable population has prompted a wave of diagnostic measures.

Diagnostic models that predict mental health and behavioral disorders are ethically paramount, making them a prevalent case study for algorithmic fairness and accountability. To explore this, we propose an audit of an ADS developed for the *Child Mind Institute, Problematic Internet Use* Kaggle competition. The purpose of this ADS is to predict a youth’s Severity Impairment Index (*sii*) by relying on physical proxy data, mainly physical activity metrics and internet usage behavior.

By auditing this system, we aim to uncover potential performance disparities across demographic subpopulations (such as age and sex) and evaluate the ethical trade-offs of misclassifying a youth’s mental health severity based on these physical proxies.

We will audit the 1st-place winning implementation made by Lennart Haupts. This solution utilizes a voted ensemble of regression models (LGBM, XGBoost, CatBoost, and ExtraTrees) to predict the continuous *PCIAT* – *PCIAT_Total* score, which is then converted into the final *sii* classification.

The pipeline has extensive data processing, including Lasso imputation for missing tabular data, PCA dimensionality reduction, and explicit feature normalization.

2 Input and Output

2.1 Input

2.1.1 Data Collection & Source

The data used by this ADS originates from the Healthy Brain Network (HBN), a large-scale initiative by the Child Mind Institute to collect biomedical data on 5,000+ children aged 5–21 in the New York City area. Participants were recruited through community referrals targeting families who had concerns about their child’s learning, attention, or psychiatric health, meaning the dataset is a clinically-referred, not population-representative, sample. Data collection occurred across multiple sessions and included clinical interviews, cognitive assessments, physical fitness evaluations, and multi-day wrist actigraphy. The specific data provided through the Kaggle competition is a pre-processed subset of the HBN, released in tabular (`train.csv/test.csv`) and time-series parquet form.

A critical structural feature of this data is that the ground-truth label, the Severity Impairment Index (*sii*), is derived from the Parent-Child Internet Addiction Test (PCIAT), which was administered only to a subset of participants. Rows without a *sii* value are dropped from training entirely, reducing the effective training set and introducing a potential selection bias — participants for whom PCIAT data exists may differ systematically from those for whom it does not.

2.1.2 Input Features

Basic Demographics Two features: `BasicDemos-Age` (continuous float, years) and `BasicDemos-Sex` (binary integer, 0/1).

Age spans approximately 5–22 years across the training set. Sex is a protected attribute. Although Basic_Demos-Sex is collected and present in the raw data, the released implementation excludes it from its final feature set (see §3.4). Fairness considerations nonetheless persist: correlated body-composition features may indirectly encode sex, and we examine this in §4.

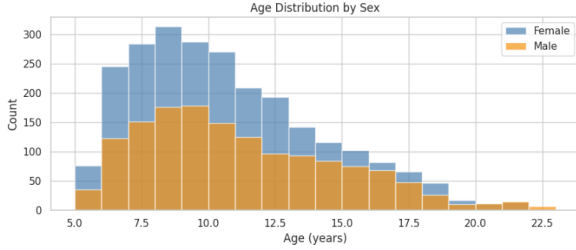


Figure 1: Age distribution by sex across the training set.

Children’s Global Assessment Scale A single numeric score (CGAS-CGAS_Score) representing a clinician-assigned global functioning rating (1–100 scale). Contains a seasonal enrollment indicator (CGAS-Season), dropped during feature engineering.

Physical Measurements Continuous measurements including height, weight, BMI, systolic/diastolic blood pressure, and resting heart rate. These are collected via direct clinical assessment. Physical-BMI and the BIA-derived BIA-BIA_BMI are both present and partially redundant; the model constructs a normalized composite (BMI_mean_norm) corrected for age-group-specific population means.

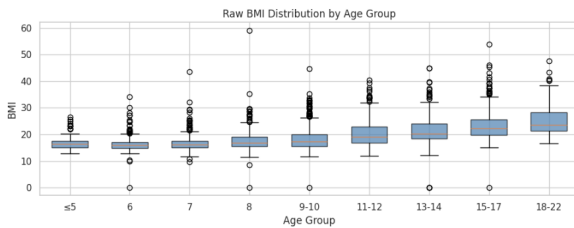


Figure 2: Raw BMI distribution by age group. Median BMI increases monotonically with age, consistent with expected pediatric growth patterns.

Fitness Endurance Captures a timed endurance test (duration in seconds, Fitness_Endurance-Time_Sec, and associated maximum heart rate). Exhibits the highest missingness of any tabular feature group, with Fitness_Endurance-Time_Sec and related

columns exceeding 80% missing in the training set, reflecting that the timed endurance test was completed by only a small subset of participants.

Fitness Gram Challenge A battery of youth fitness assessments: grip strength dominant/non-dominant hand (FGC_GSD/FGC_GSND), curl-ups (FGC_CU), push-ups (FGC_PU), trunk lift (FGC_TL), and bilateral sit-and-reach (FGC_SRL/FGC_SRR). Each measure has a corresponding ordinal Healthy Fitness Zone indicator (*_Zone). Grip values are hard-clipped to [9, 60] kg during cleaning; all zone columns are dropped after engineering age-normalized aggregate features (FGC_Zones_mean/min/max, GS_max/min). Missing rates for FGC features are moderate to high.

Bioelectrical Impedance Analysis The largest single feature group, covering body composition outputs from a bioimpedance device: fat percentage, fat-free mass (FFM), fat mass index (FMI), fat-free mass index (FFMI), basal metabolic rate (BMR), daily energy expenditure (DEE), bone mineral content (BMC), skeletal muscle mass (SMM), total body water (TBW), extracellular water (ECW), intracellular water (ICW), lean dry mass (LDM), and lean soft tissue (LST). All are continuous floats. Several exhibit implausible outliers that are nullified during cleaning (e.g., BMR > 4000 kcal, fat percentage outside [5, 60]%). The model engineers age-normalized variants (BMR_norm, DEE_norm, FFM_norm) and the ratio ICW_ECW. BIA features have among the highest missing rates of the tabular feature groups, reflecting device availability constraints at collection sites.

Physical Activity Questionnaire Self-reported activity level summary scores from either the adult proxy (PAQ_A) or child-completed (PAQ_C) versions of the Physical Activity Questionnaire. Only PAQ_A_Total is retained after feature engineering. These are ordinal-valued summary scales.

Sleep Disturbance Scale A single summary score from the Sleep Disturbance Scale for Children. Continuous integer scale.

Pre-Intervention Educational History A single feature capturing computer usage context, specifically computer use during school hours or at home (integer ordinal).

Missing Value Summary Missing data is pervasive in this dataset. Features with >30% miss-

ingness include most BIA columns and Fitness Endurance metrics. The model addresses this via a Lasso-based predictive imputer (LassoCV with 5-fold CV): for features with less than 40% missing, values are imputed using predictions from a Lasso model trained on other features; for features exceeding this threshold, mean imputation is applied. This introduces a key auditability concern — imputed values are not flagged as synthetic, meaning downstream fairness analyses cannot distinguish observed from imputed inputs.

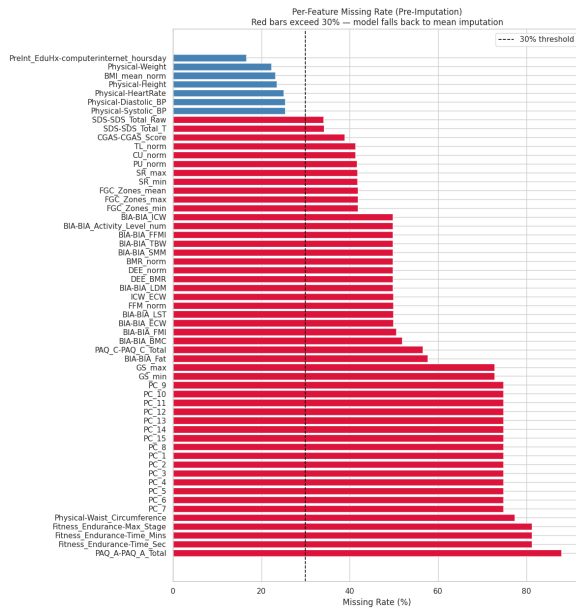


Figure 3: Per-feature missing rate (pre-imputation). The 30% line shown is illustrative; the released imputer falls back to mean imputation when feature missingness exceeds the 40% threshold ($na_frac=0.4$).

Actigraphy Time-Series Features A subset of participants wore wrist accelerometers that collected high-frequency sensor data stored as partitioned Parquet files. The raw time-series contains five channels per timestep: Euclidean Norm Minus One (enmo, a gravity-corrected movement magnitude), wrist angle (anglez), ambient light (light), battery voltage (battery_voltage), and a binary non-wear flag (non-wear_flag). Rather than training sequence models directly on the raw series, the implementation collapses the time-series into 74 summary statistics per participant via the `time_features()` function. For each of the four sensor channels, six statistics (mean, std, max, min, first-order difference mean, first-order difference std) are computed under three temporal masks: unrestricted, nighttime (22:00–05:00), and daytime

(07:00–20:00), yielding $4 \times 3 \times 6 = 72$ features, plus the mean non-wear rate and active-ENMO sum (≥ 0.05 g). These 74 features are then standardized and reduced to 15 principal components via PCA. As shown in Figure 5, PC_1 and PC_2 alone account for approximately 22% and 12% of variance respectively, with all 15 components collectively capturing approximately 82% of total actigraphy variance while controlling dimensionality. Not all participants have actigraphy data; for those without it, the PCA components are left-joined as NaN and subsequently imputed.

2.2 Output

2.2.1 Output Variable

The SII The ADS’ output is the Severity Impairment Index (*sii*). This results in an integer with a value from 0 to 3 that indicates the level of severity for an individual’s problematic internet usage.

- **0 — None:** PCIAT Total < 30
- **1 — Mild:** $30 \leq \text{PCIAT Total} < 50$
- **2 — Moderate:** $50 \leq \text{PCIAT Total} < 80$
- **3 — Severe:** PCIAT Total ≥ 80

This model does not expressly predict the value of the *sii*, but instead predicts the score of the Parent-Child Internet Addiction Test. This value is then converted to an ordinal one that fits the designated pre-determined thresholds.

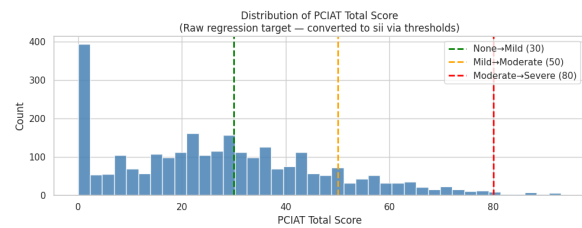


Figure 4: The distribution of PCIAT score. The dashed lines show the thresholds that are used to convert the score to *sii*.

Class Distribution There is great imbalance within the training dataset with a large emphasis on classification 0 and 1 for the *sii*. The classifications of 2 and 3 are underrepresented as the majority of participants fall into the lower severity categories. This is a risk for the accuracy of the model as it makes the model more likely to be biased toward predicting individuals as 0 or 1. We can see in

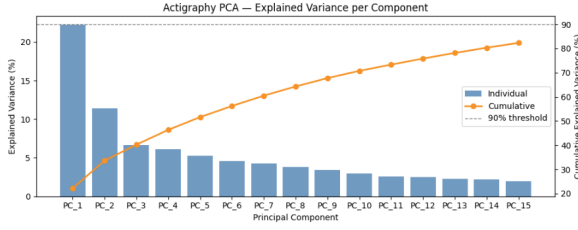


Figure 5: Explained variance per actigraphy PCA component, with cumulative variance overlay. The 15 components collectively capture approximately 82% of total variance.

Figure 6 that the distribution is extremely skewed toward the None and Mild classifications. It's essential that this is addressed so that model does not persist as biased in this area.

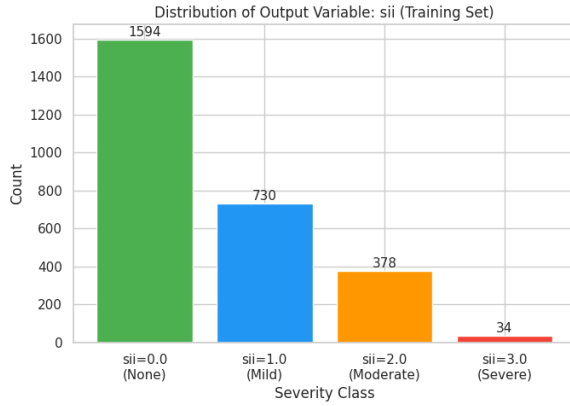


Figure 6: The distribution of the *sii*.

Evaluation Metric The Quadratic Weighted Cohen's Kappa score indicates how our model is performing on a scale of 1 to -1. This metric takes into consideration the ordinal nature of the *sii*. This means that there is a larger penalty for predictions that are further from the true value. This is in contrast to penalizing predictions based on simple accuracy, which would mean all off-predictions would have the same penalty. If a model receives This model has a 0.47 for the Quadratic Weighted Cohen's score, indicating that there is moderate-leaning agreement between the predicted values and the true values.

- **1.0** — Perfect agreement with true
- **0.6 – 1.0** — Substantial agreement
- **0.4 – 0.6** — Moderate agreement
- **0.2 – 0.4** — Fair agreement
- **0.0** — No better than random chance

- **< 0.0** — Worse than random chance

Feature Correlation As shown in Figure 7, the group-average correlation heatmap reveals that BIA features have a mean absolute intra-group correlation of approximately 0.48, reflecting their shared physiological derivation from the same bioimpedance device. BIA features also show moderate correlation (~ 0.37) with Physical measurements such as height and weight, which is expected given that body composition is partly a function of size. By contrast, the actigraphy principal components and questionnaire-based features (PAQ, SDS) show near-zero correlation with all other groups, suggesting they contribute independent signal. This high intra-BIA correlation warrants caution when interpreting tree-based feature importances, as importance may be fragmented arbitrarily across structurally redundant features.

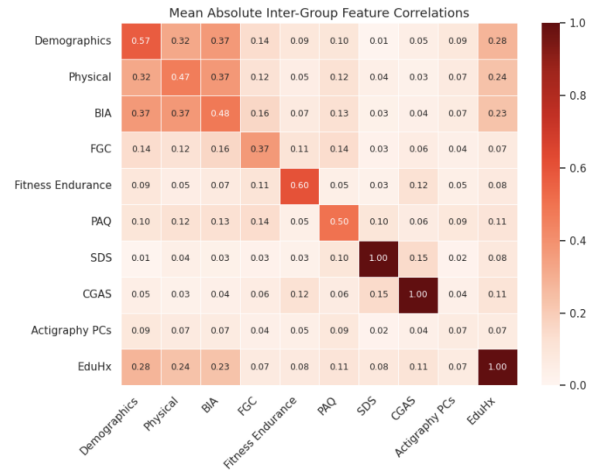


Figure 7: Mean absolute inter-group feature correlations (pre-imputation). BIA features exhibit the highest intra-group correlation (0.48), while actigraphy PCs and questionnaire features are largely independent of all other groups.

3 Implementation and Validation

We audit the publicly released solution by Lennart Haupt.¹ The pipeline regresses the continuous PCIAT-PCIAT_Total score with an ensemble of five tree-based models, then maps regression outputs to the ordinal *sii* via thresholds optimized per fold. All choices below are taken from the released notebook; we re-ran it end-to-end to verify reproducibility.

¹<https://www.kaggle.com/competitions/child-mind-institute-problematic-internet-use/discussion/>

3.1 Data Cleaning and Preprocessing

Label-conditional row drop. Rows where the target `sii` is missing are removed from training (`train[train["sii"].notna()]`). This drops $\sim 31\%$ of HBN rows and silently restricts the training distribution to the PCIAT-administered subset, preserving the selection bias documented in §2.

Outlier nullification. The `clean_features()` function hard-codes plausibility windows on the BIA-derived columns and replaces out-of-range values with NaN (later imputed): $BIA - BIA_BMR > 4000$ kcal, $BIA - BIA_DEE > 8000$ kcal, $BIA - BIA_Fat \notin [5, 60]\%$, $BIA - BIA_BMC \notin (0, 10]$, $BIA - BIA_FFM \notin (0, 300]$, $BIA - BIA_FMI < 0$, and similar caps on ECW, LDM, LST, SMM, TBW. Grip strength ($FGC - FGC_GSND/GSD$) is hard-clipped to $[9, 60]$ kg rather than nullified. The thresholds are not justified in the source; they appear to be eyeballed from the training distribution.

Column drops. All ten `*-Season` columns and the 21 per-question PCIAT items (`PCIAT-PCIAT_01...20` plus `PCIAT-Season`) are excluded from the feature set. The latter is necessary: the per-item scores deterministically sum to `PCIAT-PCIAT_Total`, which is the regression target.

3.2 Feature Engineering

Age binning. An ordinal age-group variable is created via `assign_group()` using thresholds $[5, 6, 7, 8, 10, 12, 14, 17, 22]$ years, producing 9 bins. This bin index is the basis for nearly all engineered features.

Age-normalized features. Eight feature families are normalized by age-group population means encoded as hard-coded dictionaries inside `feature_engineering()`:

- `BMI_mean_norm`: arithmetic mean of `Physical-BMI` and `BIA-BIA_BMI` divided by age-group BMI;
- `GS_max`, `GS_min`: row-wise max/min of dominant/non-dominant grip strength, divided by age-group reference;
- `CU_norm`, `PU_norm`, `TL_norm`: curl-ups, push-ups, trunk lifts divided by age-group reference;
- `BMR_norm`, `DEE_norm`: BIA energy expenditure divided by age-group reference;

- `FFM_norm`: fat-free mass divided by age-group reference.

The reference values are unsourced constants in the notebook; we could not locate a citation to a pediatric reference standard. From an audit standpoint, these constants encode an implicit population assumption that travels with the model.

Aggregates and ratios.

`FGC_Zones_mean/min/max` aggregate the seven zone columns; `SR_min/max` aggregate left/right sit-and-reach; `DEE_BMR` stores the additive difference $DEE - BMR$; `ICW_ECW` stores the ratio ECW/ICW (the variable name is reversed from its definition in the code, which is a mild auditability concern). The 19 source columns feeding these aggregates are then dropped.

Quantile binning. Fifteen continuous features (e.g., `PAQ_A-PAQ_A_Total`, `BMR_norm`, `DEE_norm`, `Physical-HeartRate`, `Fitness_Endurance-Time_Sec`) are discretized into 10 quantile bins via `bin_data()`. Crucially, bin edges are computed on the *concatenation of train and test*, which leaks marginal-distribution information from test into training. The discretization also discards within-bin signal, which is acceptable for tree learners but not loss-free.

Actigraphy summarization.

The `time_features()` function reduces each participant's raw multi-day Parquet series to 74 summary statistics: for each of the four channels (`enmo`, `anglez`, `light`, `battery_voltage`), six statistics (mean, std, max, min, first-difference mean, first-difference std) are computed under three temporal masks (unrestricted, night 22:00–05:00, day 07:00–20:00), yielding $4 \times 3 \times 6 = 72$ features, plus the mean non-wear flag and active-ENMO sum ($enmo \geq 0.05$ g). Series are loaded in parallel via `ThreadPoolExecutor`. The 74-vector is standardized with `StandardScaler`, NaNs are filled with the (post-scaling) column mean, then reduced to 15 PCs via `sklearn.decomposition.PCA`. Participants without actigraphy receive NaN-filled PCs, joined back left-onto the tabular table and imputed downstream.

3.3 Lasso-based Imputation

The `Impute_With_Model` class implements predictive imputation. For each feature with >0 missing values, a clone of `LassoCV(cv=5)` is fit on rows where that feature is observed, using the remaining

features (each pre-filled with their training mean) as predictors. The class restricts predictor selection to columns whose missingness on the target’s missing rows is at most `na_frac=0.4`; for features whose own missingness exceeds this threshold, mean imputation is used instead (`"mean", np.mean(...)`). Critically, the imputer is fit on the training data and applied to both train and test, but *imputed cells are not flagged*. Any subsequent fairness or robustness analysis cannot tell observed from synthetic inputs.

3.4 Model Implementation

Final feature set. After engineering and binning, an additional 21 features are removed via a hard-coded exclude list, yielding the `reduced_features` set. Notably, `Basic_Demos-Sex` appears in this list and is therefore *dropped from the model entirely*. This corrects the impression given by §2.1.2: although sex is collected and present in the raw table, it is not a direct input to the released ensemble. Indirect encoding through correlated body-composition features remains possible and is examined in §4. Eight of the 15 actigraphy PCs are also dropped here, leaving 7 PCs in the final feature matrix.

Ensemble. Five regressors are trained, all targeting the continuous `PCIAT-PCIAT_Total`: an `LGBMRegressor` with Poisson objective, two `XGBRegressor`s with Tweedie objective and distinct hyperparameter sets, a `CatBoostRegressor` with RMSE loss, and an `ExtraTreesRegressor`. Hyperparameters are hard-coded constants in the notebook from a prior Optuna search; `optimize_params=False` disables re-tuning by default, so we audit the static configuration that was actually submitted.

Sample weighting. `calculate_weights()` discretizes `PCIAT-PCIAT_Total` into 10 equal-width bins and assigns inverse-frequency weights (normalized to mean 1). All five base models are trained with these weights, partially compensating for the heavy zero-inflation visible in Figure 4.

Cross-validation and threshold optimization. `StratifiedKFold` with `n_splits=10` stratifies on the ordinal `sii` while training on the continuous `PCIAT-PCIAT_Total` — a deliberate mismatch to keep class balance across folds despite the regression target. Within each fold, the trained model predicts the training-

fold scores, and `optimize_thresholds()` runs `scipy.optimize.minimize` (Powell) over a 3-vector of cut-points, with $-QWK$ as the objective and the clinical thresholds `[30, 50, 80]` as the warm start. The optimized thresholds are stored per fold; the per-model ensemble threshold is the mean across folds (`*_thresholds_ens`). The final cut-points used at inference therefore differ from the clinical PCIAT thresholds, are model-specific, and are tuned on QWK. This is a meaningful audit finding: the ordinal label produced by the system is no longer derived from the published PCIAT clinical interpretation.

Aggregation. OOF and test predictions from each base model are first thresholded using that model’s ensemble thresholds, producing five per-row ordinal predictions. These are aggregated via `scipy.stats.mode` (hard plurality voting) when `voting=True` (the default). An alternative weighted-average path exists with hand-tuned weights `[0.2, 0.2, 0.3, 0.1, 0.2]` but is not used in the submitted configuration. A `LinearRegression` meta-model is fit purely for diagnostic inspection of base-learner coefficients and does not participate in the final prediction.

3.5 Validation as Performed by the Author

The author reports per-fold QWK and $\text{mean} \pm \text{std}$ across the 10 folds for each base model, an OOF QWK on the voted ensemble (the headline number is ≈ 0.47), the OOF confusion matrix, and a sensitivity sweep that fixes thresholds 2 and 3 at `{50, 80}` and varies threshold 1, demonstrating that QWK is steeply sensitive to the first cut-point. We replicate these numbers exactly with the released seed (`SEED=9365`). The author does not report subgroup performance, calibration of the regression layer, robustness to imputation, or seed variance — gaps that motivate the analyses in §4.

4 Outcomes

4.1 Accuracy Analysis:

In order to evaluate the accuracy of the model with the Quadratic Weighted Kappa (QWK). The QWK by design punishes values that veer further from the true value than others. This is so that nearby misclassifications are not weighted the same as ones that are further off. We use the the out-of-fold (OOF) ensemble predictions to find estimates of performance. We choose QWK because the *sii* is ordinal by nature. There is a major difference

between classifying a truly Severe case as None, than classifying a truly Severe case as Mild.

Accuracy of Sex We see that there is a QWK of 0.460 for female participants and a QWK of 0.470 for male participants. This shows us that there is not a meaningful difference between gender features. There are not a structurally different errors across genders.

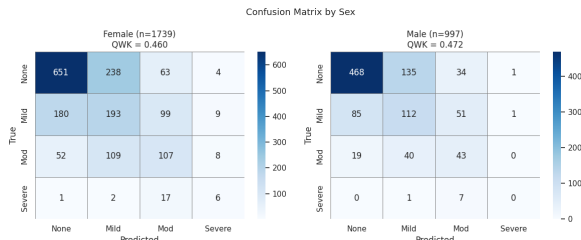


Figure 8: These are confusion matrices by sex. Female ($n = 1739$, QWK = 0.460) and male ($n = 997$, QWK = 0.474) participants have structurally similar error patterns.

Accuracy of Age Groups There are extremely different outcomes for the different age groups. We see the highest performance for the age group that houses those aged 11 to 12 years old. On extreme values of age, we see that there is low performance. For example, there is a QWK of 0 for children 5 years old and younger, a QWK of 0.018 for children aged 6, and a QWK of 0.099 for participants between 18 to 22 years old. We also note that the most represented age groups in the training data are those that house 11 to 12 year olds and 9 to 10 year olds respectively. No group achieves the average QWK of 0.47, indicating the model's inability to highlight intra-group performance failures. This is likely due to the training set being dominated by middle centered age groups, making the behaviors of that population the default baseline for predictions.

4.2 Fairness Analysis:

We analyze fairness using the direction of systematic prediction error with mean prediction error, and clinical error rates with FNR and FPR. We define a binary positive label as $s_{ii} \geq 2$, meaning there is either a moderate or severe case. We do this primarily for clinical error rate analysis. We choose FNR and FPR due to the failure to identify a child with need is more harmful than creating a false alarm. This justifies the selection of a recall based fairness

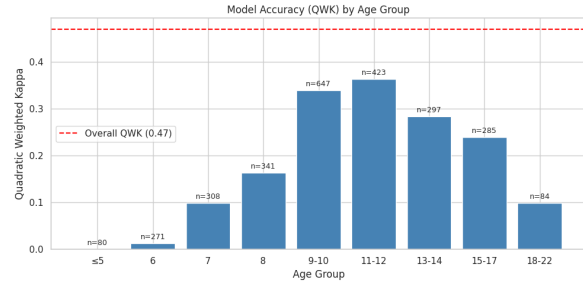


Figure 9: A model accuracy, QWK, by age group. The dashed red line shows the overall QWK of 0.47. No individual age group reaches this QWK threshold.

emetric. False positives in this case would lead to an in-need child going without medical intervention.

Mean Prediction Error The mean prediction error uses the direction of the errors to determine systematic bias. A positive value tells us that the model is overestimating severity, and a negative value tells us that the model underestimates severity.

By age group, we see a clinically significant pattern arises (Figure 10). We see that the model overestimates severity for both male and female participants. There is a female mean error of 0.048 and male mean error of 0.088. Thus, there is not a meaningful disparity by sex for this dataset.

When we look at age groups, we see that the model heavily underestimates severity for certain ages (Figure 10). In particular, younger age groups have mean errors between -0.124 and -0.231. In contrast, participants in older age groups face overestimation by the model. For example, age groups for both 11–12 and 18–22 year old participants have mean errors between 0.303 and 0.702. Given the skew for age of the training data, we see that the model might have learned that the behaviors middle-aged participants are the central tendency, and makes predictions for other age groups with this in consideration.

False Negative and False Positive Rates When we consider sex, we see that FNR is 0.540 and 0.536 for female and male participants respectively. FPR for females is 0.122 and for males is 0.098. This is not a meaningful difference and shows that there is not a significant difference of fairness between the sexes.

There is a significantly different FNR and FPR outcome across the different age groups. Primarily, the model has an FNR of 1 for children between 5–8 years old. Thus, the model is unable to flag

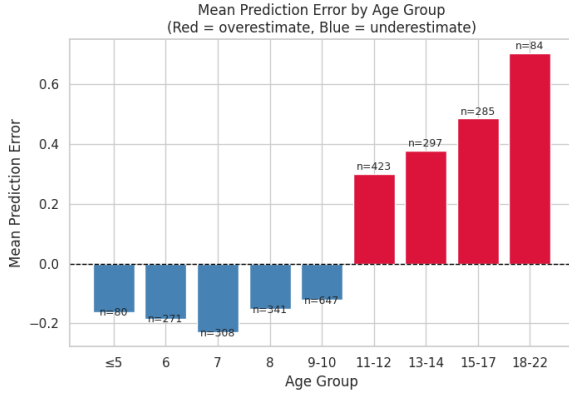


Figure 10: The mean prediction error (predicted — true) by age group. Younger groups are systematically underestimated and older groups are systematically overestimated.

participants from this group as moderate and severe when necessary. This is similar for participants between the ages of 9–10, with an FNR of 0.932. On the other hand, older participants have extremely high FPR values. We see FPR values of 0.351, 0.411, and 0.592 for participants of ages 13–14, 15–27, and 18–22 respectively.

The model effectively ignores young participants that are in need of attention, and flags severity for older participants that are otherwise without an issue.

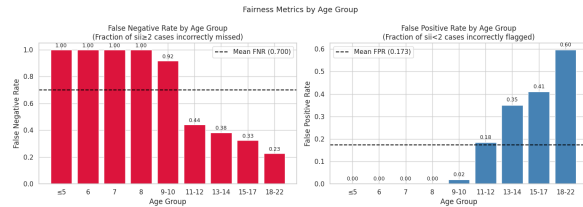


Figure 11: False negative rate (left) and false positive rate (right) by age group, with positive defined as $sii \geq 2$. FNR = 1.000 for children aged 5 to 8. FPR = 0.597 for participants aged 18 to 22.

4.3 Robustness and Stability:

We will determine the robustness of the model by looking at how sensitive it is to missing values. For example, we will look at whether or not performance of the model lowers when there are missing values. We will also cross-examine the model with different random seeds.

Performance on Rare Classes We see that the model’s performance decreases consistently with increasing severity. The model is able to correctly and accurately identify 70.2% of None cases ($sii =$

0, $n = 1594$), 41.5% of Mild cases ($n = 730$), 40.5% of Moderate cases ($n = 378$), and 17.6% of Severe cases ($n = 34$). All in all, only 6 of the 34 actual Severe cases are correctly identified by the model.

The model tends to over predict Mild cases with 826 predicted compared to 730 true, and under predicts Severe cases with 28 predicted compared to 34 true. This shows that the effect of the skewed class distribution within the training data, and the sample weighting strategy have a combined effect. This effect does correct for class imbalance somewhat but is not able to eliminate the bias toward prediction on the low severity end of the spectrum. Therefore, given the recall of 0.176 for severe participant cases, we can consider this to be a failure in terms of reliability.

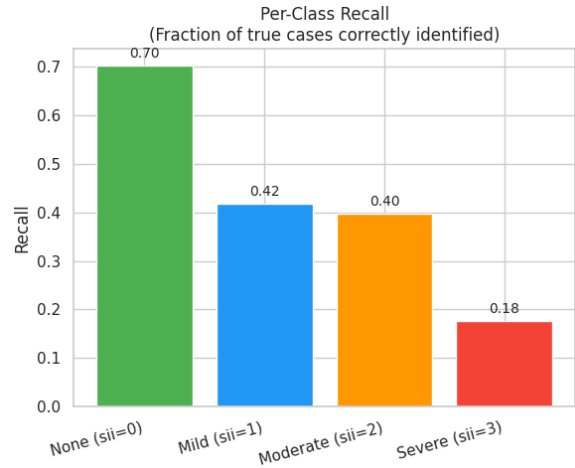


Figure 12: The per-class recall (right) and true vs. predicted class counts (left). Recall degrades consistently with severity. Reaches 0.176 for severe cases ($sii = 3$).

Missing Data Sensitivity We decided to partition the training samples into quartiles by the number of missing raw features prior to imputation in order to determine robustness. We also computed QWK for each quartile. Our results reveal that Q1 reached a QWK of 0.426, Q2 reached a QWK of 0.461, Q3 reached a QWK of 0.484, and Q4 reached a QWK of 0.432. We also see that Q1 had an average of 2.7 missing features with $n = 687$, Q2 had 5.3 missing with $n = 926$, Q3 had 12.9 missing with $n = 670$, and Q4 had 19.3 missing with $n = 453$.

These results show us that performance does not consistently decrease alongside missingness. For example, Q3 reaches the highest QWK out of all of the other quartiles. Likewise, we see that the total range for the quartiles is around 0.06. Therefore,

the Lasso based imputation strategy performs decently well. Missing data does not appear to be a significant source of model unreliability. We state this with caution, as imputed values are not flagged in our dataset. This means that the analysis cannot differentiate missingness cases that were concentrated in low signal features from cases where imputation lead to actually accurate predictions.

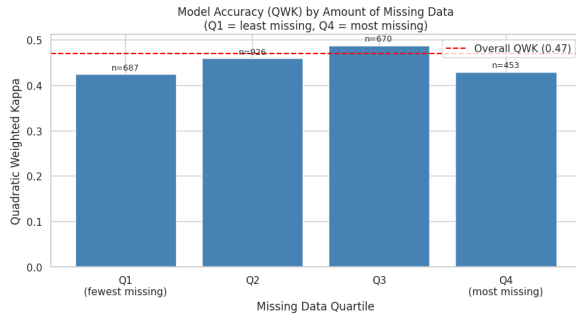


Figure 13: Model accuracy, QWK, by missing data quartile. Performance does not degrade consistently with missingness. This suggests that the Lasso imputation strategy is effective.

Seed Stability We decide to assess whether or not the model is sensitive to random initialization. We do this by re-running the LGM base model with five alternative random seeds, and then recording the QWK values that come out as output. We highlight seeds 1234, 2024, 4242, 7777, and 9999. For these seeds there are QWK values from 0.450 to 0.470, a mean of 0.436, and a standard deviation of 0.007. This is low variance and suggests that the model has a relatively stable performance that is not the result of one particularly good seed. The original seed of 9365 from the originally submitted model is able to produce outcomes that are representative of typical behaviors across initialization.

5 Summary

Data Appropriateness There are several structural limitations that are introduced within the Healthy Brain Network dataset, which affect the performance of the selected ADS. Most significantly, the sample is clinically-referred and is therefore not representative of the population. Participants within the study were referred by their parental guardians due to health concerns. This means that the training data involves a sample that is inherently different or abnormal from the general population of youths. Thus, children that do not have certain behavioral or physical concerns by

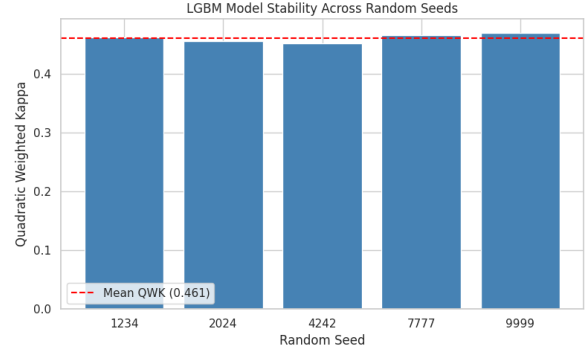


Figure 14: The LGBM model QWK across five random seeds. Performance ranges from 0.450 to 0.470 with a standard deviation of 0.007. This confirms stable initialization.

their parents are not included in the model's analysis. Due to this, the model's feature associations potentially do not generalize to the population of the study.

Additionally, the ground truth label, the *sii*, is a feature of the Parent Child Internet Addiction Test. This test was only given to a small amount of the participants. Any rows without a *sii* value are dropped from the dataset. This reduces the training set by around 31% and also introduces selection bias. There may be systematic differences between participants with PCIAT data and those without. This is not taken into account by the model.

Lastly, the selected ADS is dependent on physical proxy data features such as composition, fitness assessments, and actigraphy in order to make its predictions of internet usage. There is not a clear causal relationship between physical attributes and problematic internet usage. While the training set may reveal certain associations between these two features, this is likely mediated by age. There are different values for fitness, sleep patterns, and physical developments for young children compared to young adults. The extreme age related disparities are consistent with the model having been trained to predict age rather than internet behaviors. Therefore, construct validity was introduced to the model because of the reliance on physical proxy data. In other words, the issues with the model stem from the training dataset more so than the actual model's build.

Robustness, Accuracy, and Fairness The model has an overall QWK of 0.47, which indicates moderate performance. However, there are many discrepancies within the predictions once we focus on

certain subgroups. No age group achieves a score of 0.47, and participants aged 5 have an overall QWK of 0. Essentially, the model is unhelpful for signal and predictive ability for younger children.

We calculated FNR across age groups to determine instances where the model missed potentially severe cases. For participants aged 5 to 8 there was a FNR of 1.0, indicating the model missed identifying every truly severe case in the sample. The FPR for 18 to 20 year olds was 0.597, again indicating that the model over predicted severity for older participants. There are no meaningful sex differences.

For robustness, we found that the model was stable across all random seeds. The model was also very resilient toward missing data and had variation with the QWK by at most 0.06 across missingness quartiles. The instability of the outcomes mainly comes from structural problems such as age and class imbalances within the training data. This is what lead to the severe age related discrepancies.

Stakeholders for this ADS include clinicians, school counselors, and parents. A missed severe case for a young child would be of extreme importance to these groups. Likewise, it would be harmful for reputation to flag a child that is doing fine. Parents would also be alarmed at false positives. This makes FNR and severe-case recall are the most clinically relevant. These groups would use all the predicted outcomes of the model to determine which children need more time and resources allocated to them for their internet usage. The most relevant metrics for these groups are recall for severe cases and FNR for young participants. Given these standards, the model is not clinically appropriate or sufficient. In particular, the model tends to miss severity within young children, which are an extremely vulnerable and sensitive population. This inability to predict correct severity would lead to children with problematic internet usage remaining at risk.

Deployment Considerations All in all, we believe that this ADS should not be clinically implemented as it is right now. An FNR of 1.0 for very young children is extreme and unacceptable, and leads to extremely low diagnostic value. Additionally, this type of model could lead to distrust of clinical providers if used in real life settings. There are also concerns for unsourced normalization constants in `feature_engineering()`, test-set leakage in `bin_data()`, unflagged imputed values, and

thresholds that are re-optimizes away from the published PCIAT clinical cutpoints. This means that the output *sii* does not correspond to its clinical interpretation anymore.

Recommendations For future studies, we recommend a few points to consider.

For *data collection*, there should be an aim to include a sample that is representative of the population. The sample should not be only clinically-referred, and should have PCIAT values for all enrolled participants. This would prevent selection bias from entering the training dataset. The model would benefit from the oversampling of young participants and participants with moderate and severe *sii* scores. This would prevent age and class imbalances that lead to the model being an overall failure for predictions.

For *feature engineering*, age group normalization constants should be sourced from and cited against pediatric reference standards. This is in contrast to deriving the constants from the training distribution. The ICW_ECW variable should also be changed to match its designated definition. In order to prevent test set leakage, quantile bins should be computed on training data rather than the combined train and test set.

For *model design*, the model should be trained using distinctive age bands. Otherwise, age should be considered a stratification variable with explicit performance constraints per stratum. These suggestions would help correct the severe age-group balances seen in the outcomes. Likewise, class imbalance should be addressed with oversampling of the minority groups, or by asymmetric loss functions that penalize severity-miss more than mild-miss. Lastly, imputed values should be flagged with a binary indicator. This would allow clinicians to be able to discern observed inputs from synthetic inputs.

For *validation and reporting*, metrics such as FNR, FPR, and QWK should be reported as a study standard. The regression layer should also be assessed based on its calibration to establish the calibration of the PCIAT as more than ordinal thresholds. In order to avoid clinical misinterpretation, certain optimized thresholds should be documented and be easily discerned from the clinical PCIAT thresholds. These implementations mainly relate to transparency, and allow stakeholders to have a better understanding of how the model determines outcomes.